

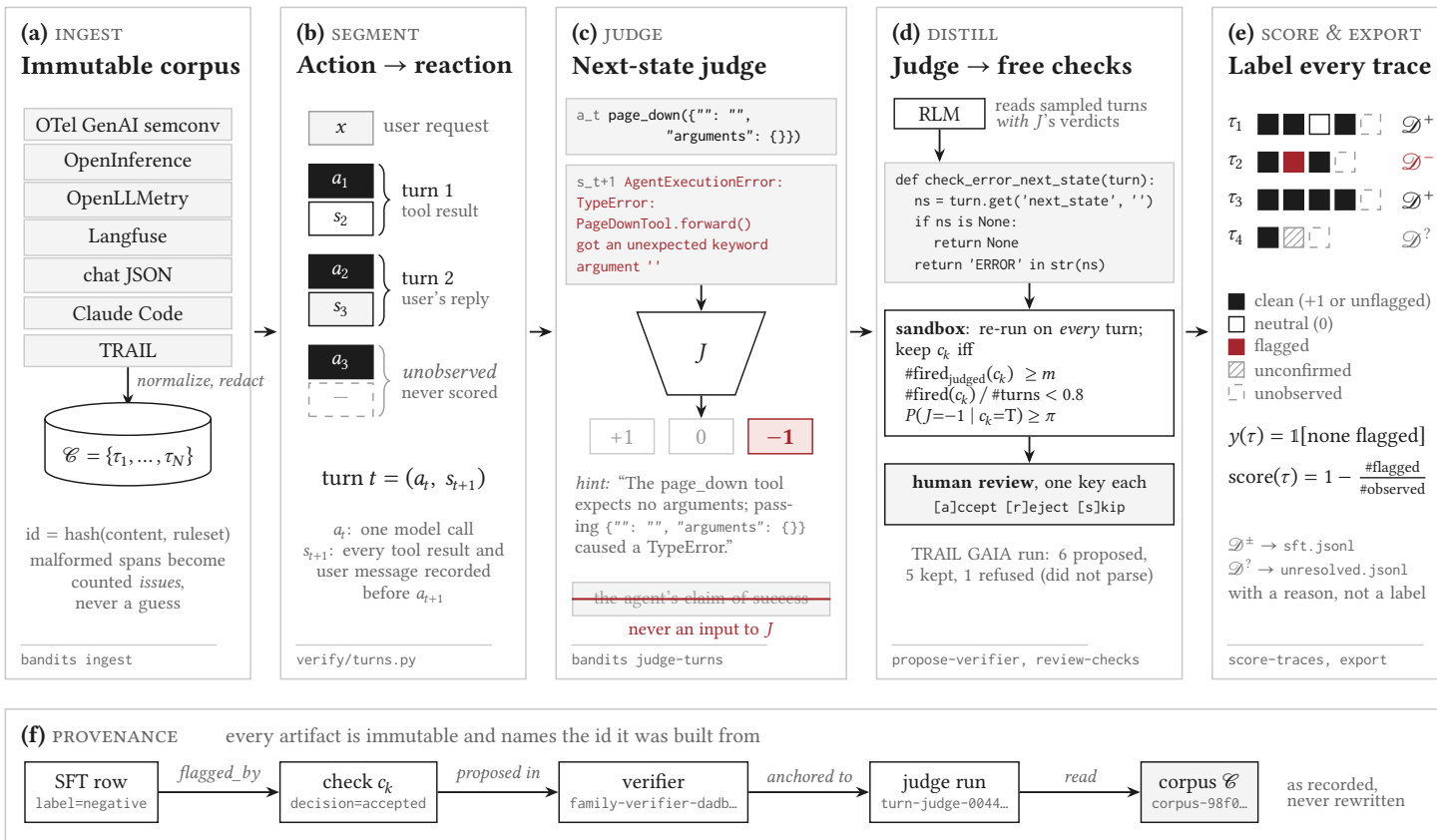


Figure 1. Method. (a) Traces in any supported format are normalized into a content-addressed corpus. (b) Each model call a_t is paired with its reaction s_{t+1} : the tool results and user messages recorded before the next call. (c) A judge scores each observed turn from the reaction alone. The agent’s account of its own success is never an input. (d) An RLM distills the judge’s verdicts into plain predicates, each re-executed over every turn and kept only if it agrees with the judge. A human then accepts or rejects it, so scoring a new trace costs no model call. (e) A trace with a flagged turn is a negative, a clean one a positive, and one that cannot be confirmed either way is quarantined. (f) Any exported row resolves back to the corpus it came from. The action, reaction and hint in (c) are verbatim from a TRAIL GAIA run, and the check in (d) is the one the RLM wrote there, shortened to an equivalent form.